

Day 26 – Screening Scoring Engine

1. Objective

The Screening Scoring Engine evaluates candidate responses from the screening workflow using multiple assessment criteria.

The engine calculates structured screening scores based on:

- Clarity
- Relevance
- Completeness
- Consistency

The scoring system converts individual criterion scores into normalized values and produces an explainable final screening score.

2. Processing Capabilities

The Day 26 Screening Scoring Engine supports:

- Screening response scoring
- Clarity evaluation
- Relevance evaluation
- Completeness evaluation
- Consistency evaluation
- Score normalization
- Score boundary handling
- Per-question scoring
- Multiple-question screening evaluation
- Final screening score calculation
- Explainable scoring output
- Empty screening handling

3. Screening Scoring Flow

Candidate Screening Response | v Criterion Scores | +-----+ || v v v Clarity Relevance
Completeness || | +-----+-----+ | v Consistency | v Score Normalization | v Weighted Score | v
Final Screening Score

4. Scoring Criteria

The engine evaluates four primary criteria:

Clarity

Measures how clearly the candidate communicates the answer.

Relevance

Measures how closely the response addresses the screening question.

Completeness

Measures whether the candidate provides sufficient information in the response.

Consistency

Measures the consistency and reliability of the candidate's response.

5. Score Normalization

Individual criterion scores are accepted on a 0–10 scale.

The engine normalizes these values into a 0–100 scale before applying the configured criterion weights.

Scores outside the supported range are safely constrained to the valid range.

Examples:

- 0 → 0
- 5 → 50
- 10 → 100

6. Per-Question Scoring

The engine provides a `score_question()` method for evaluating an individual screening response.

The result contains:

- Question ID
- Individual criterion scores
- Normalized weighted score
- Score-weight explanation

This provides an explainable representation of how each screening response was evaluated.

7. Multiple-Question Scoring

The `score_screening()` method accepts multiple screening responses.

Each response is evaluated individually before being combined into the final screening result.

The final screening score represents the average of the normalized per-question scores.

8. Structured Output

The final screening result contains:

- Total number of questions
- Final screening score

- Individual question scores
- Explanation of the final score calculation

This structured output can be used by downstream screening and candidate evaluation components.

9. Empty Screening Handling

If no screening responses are provided, the engine safely returns:

- Total questions: 0
- Final screening score: 0.0
- Empty question-score list
- Explanation indicating that no responses were provided

10. Explainability

The scoring engine provides weight information for each criterion:

- Clarity weight
- Relevance weight
- Completeness weight
- Consistency weight

This allows downstream components to understand how the final score was produced.

11. Implementation

Main implementation:

`scoring/screening_scoring_engine.py`

Configuration:

`data/screening_scoring_configuration.json`

Automated tests:

`tests/test_screening_scoring_engine.py`

12. Testing

The Day 26 automated test suite validates:

- Configuration file availability
- Score normalization
- Score boundary handling
- Individual question scoring
- Perfect score calculation
- Multiple-question scoring
- Perfect screening evaluation
- Empty screening handling
- Explainable scoring output

Test Result

All Day 26 Screening Scoring Engine tests passed successfully.

9 passed in 0.09s

13. Day 26 Deliverable

The completed Screening Scoring Engine provides a structured and explainable scoring layer for candidate screening.

The engine can evaluate individual screening responses, aggregate multiple responses, normalize scores, and produce a final screening score suitable for downstream candidate evaluation.